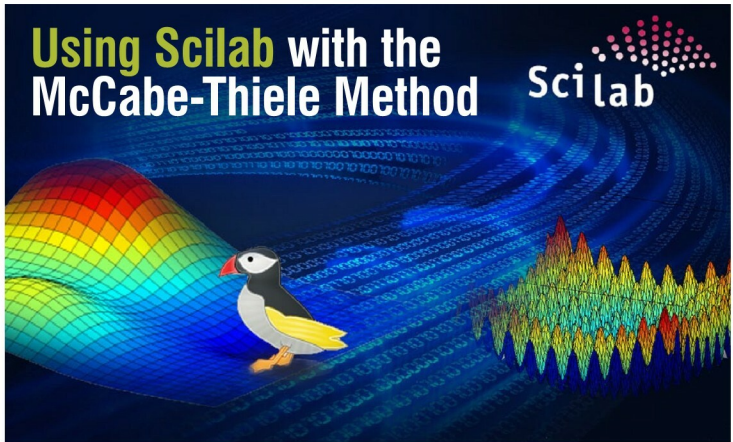




Cross-platform, open source numerical computational package, Scilab, can be used for image enhancement, fluid dynamics simulations and numerical optimisation, among other things. When used with the McCabe-Thiele method, which is said to be the simplest and perhaps most instructive method for the analysis of binary distillation, the combination is formidable.

The McCabe-Thiele method was published 90 years ago by Warren L. McCabe and Ernest W. Thiele in an article titled 'Graphical Design of Fractionating Columns'. To use their own words, "... the method consists essentially in drawing, on the same rectangular plot, the equilibrium curve for vapour and liquid compositions and straight lines representing the equations for enrichment from plate to plate, and passing from one to another in a series of steps." Scilab is an open source software for numerical computation with a syntax similar to MathWorks' MATLAB. The aim of this article is to show how Scilab can be used to plot the McCabe-Thiele diagram and simulate the distillation of a binary mixture. All the Scilab examples presented here are made under Linux Mint 17 Xfce.

### The vapour-liquid equilibrium curve

First, it's necessary to plot the vapour-liquid equilibrium (VLE) curve for the most volatile (lower-boiling) component in the mixture. For the two mixtures discussed here (ethanol/water and chloroform/methanol), the data points (x,y) are taken from the third reference given at the end of this article. Note that the point at which the VLE curve cuts the diagram on the

diagonal is called the azeotrope. At this point, the vapour phase has the same composition as the liquid phase and the mixture cannot be separated by a simple distillation. The data points are now fitted with three different techniques to obtain the VLE curve equation:

- Cubic spline monotone
- Rational function 1 (with four coefficients)
- Rational function 2 (with six coefficients)

```
// Rational function 1
deff("[y]=f(a,b,c,d,x)", "y=((a+b*x)./(1+c*x+d*x.^2))");
deff("[e]=res(A,x,y)", "e=f(A(1),A(2),A(3),A(4),x)-y");
Ainit=[1;1;1;1];
[S,coe]=leastsq(list(res,x,y),Ainit);
yp1=f(coe(1),coe(2),coe(3),coe(4),xx);
```

```
// Rational function 2
deff("[y]=f(a,b,c,d,e,f,x)", "y=((a+b*x+c*x.^2)./(1+d*x+e*x.^2+f*x.^3))");
deff("[e]=res(A,x,y)", "e=f(A(1),A(2),A(3),A(4),A(5),A(6),x)-y");
Ainit=[1;1;1;1;1;1];
[S,coe]=leastsq(list(res,x,y),Ainit);
yp2=f(coe(1),coe(2),coe(3),coe(4),coe(5),coe(6),xx);
```